• **DRIVEABILITY—MALFUNCTION INDICATOR LAMP (MIL) ILLUMINATED—DIAGNOSTIC TROUBLE CODES (DTC) P0300/P0301 STORED IN MEMORY—VEHICLES EQUIPPED WITH 4.0L SOHC ENGINE ONLY** • **ELECTRICAL—LAMP—MALFUNCTION INDICATOR LAMP (MIL) ILLUMINATED—DIAGNOSTIC TROUBLE CODES (DTC) P0300/P0301 STORED IN MEMORY—VEHICLES EQUIPPED WITH 4.0L SOHC ENGINE ONLY** • **ENGINE—4.0L SOHC—MALFUNCTION INDICATOR LAMP (MIL) ILLUMINATED—DIAGNOSTIC TROUBLE CODES (DTC) P0300/P0301 STORED IN MEMORY—VEHICLES EQUIPPED WITH 4.0L SOHC ENGINE ONLY**	**Article No. 02-3-2**

FORD: 1997-2002 EXPLORER SPORT, EXPLORER
2001-2002 EXPLORER SPORT TRAC, RANGER

MERCURY: 1997-2002 MOUNTAINEER

ISSUE

Some vehicles equipped with the 4.0L SOHC engine may exhibit a CHECK ENGINE LIGHT "ON" with Diagnostic Trouble Codes (DTC) P0300 (Random Misfire) or P0301. The DTC will set when the engine is operated at high RPM'S (4800 RPM and higher). The Misfire code results due to system interaction of higher than expected transient exhaust system backpressure (under heavy acceleration) coupled with valve spring resonance points experienced at these engine speeds. This interaction may cause the exhaust valve to hang open at engine speeds of 4800 RPM and higher.

ACTION

Verify that the DTC sets only at 4800 RPM or higher. Remove current intake and exhaust valve springs and replace with new service intake and exhaust valve springs. Refer to the following Service Procedure for details.

SERVICE PROCEDURE

NOTE
THIS REPAIR ONLY APPLIES TO P0300/P0301 DTC'S SET ABOVE 4800 RPM. THIS REPAIR WILL NOT CORRECT MISFIRE DTC'S SET AT LOWER RPM'S.

NOTE
PRIOR TO REPAIR, REFERENCE YOUR APPLICABLE WORKSHOP MANUAL FOR CORRECT VALVE SPRING REMOVAL/REPLACEMENT PROCEDURES.

1. Gain access to the valve springs.
2. Remove (quantity 12) existing intake and exhaust valve springs F77Z-6513-AA and replace with new service intake and exhaust valve springs 2L2Z-6513-AA.

NOTE
ALL "12" VALVE SPRINGS ("6" INTAKE AND "6" EXHAUST) MUST BE REPLACED IN ORDER TO MAINTAIN/ENSURE PROPER VALVETRAIN DYNAMICS.

3. Road test vehicle to verify repair.

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Article No. 02-3-2 Cont'd.

LABOR OPERATION CLAIMING CHART			
Operation	**Labor Description**	**Vehicle**	**Time**
020302A	Replace All Intake And Exhaust Valve Springs	1998-2001 Mountaineer -	6.1
		1997-2001 Explorer -	6.1
		2001 Sport Trac -	6.1
		2001 Ranger -	6.6
		2002 Ranger, Sport Trac -	6.6
		2002 Explorer, Mountaineer -	5.4

PART NUMBER	PART NAME
2L2Z-6513-AA	Valve Spring - Intake And Exhaust

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage

DEALER CODING

BASIC PART NO.	CONDITION CODE
6513	42

OASIS CODES: 499000, 608000, 609000, 610000, 611000, 611500, 698298